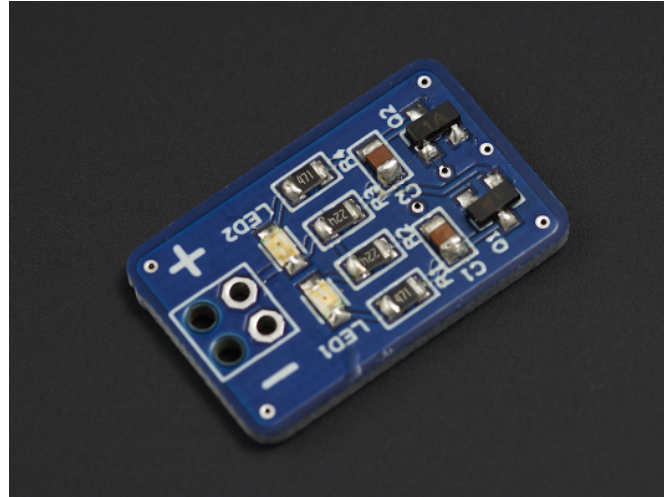

LED ALTERNATING INDICATOR SMD



INSTRUCTIONS

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Contents

1	Basics	2
1.1	Soldering	2
1.2	Resistors	2
1.2.1	Exercises	3
1.3	Soldering LEDs	4
1.3.1	Exercises	5
2	LED alternating indicator SMD	6
2.1	Bill of Materials	6
2.2	Circuit Diagram	6

1 Basics

1.1 Soldering

Our soldering exercises are designed to teach you the basics of soldering step by step. Don't worry, no one becomes an expert overnight! Just like in many things, *"practice makes perfect."*

You can find a helpful video about common soldering mistakes on our YouTube channel:

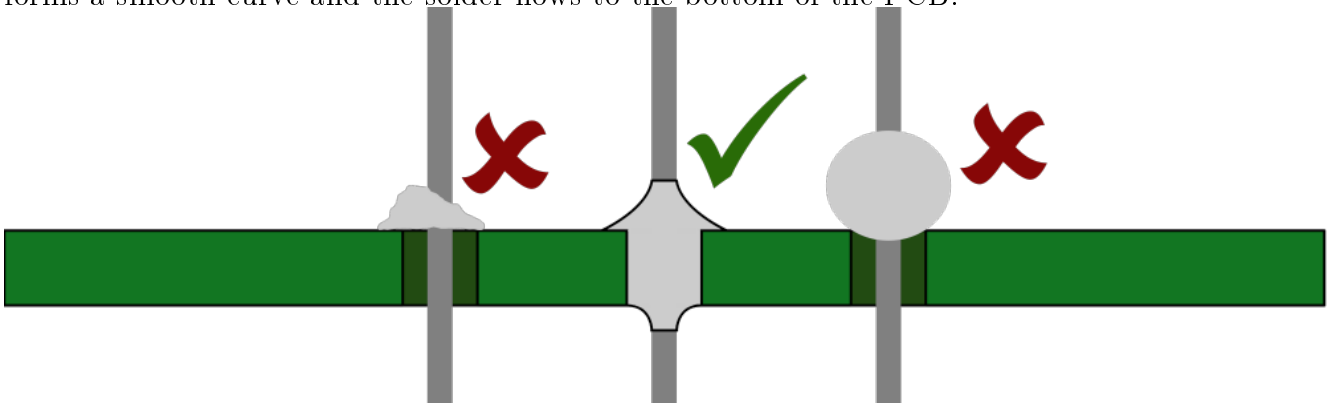
<https://youtu.be/CPXZM8r8xFw>

Here are some important points in brief:

- The ideal **temperature** for the soldering iron should be around **350°C**. It is recommended to use lead-free solder.
- The choice of soldering tip should match the size of the component. A **1.6 mm chisel tip** is ideal for most solder joints – it works for SMD components of size 0805 and wires up to 1.5 mm².
- To protect yourself from solder splashes or flying wires when cutting, it's recommended to wear **safety glasses**.
- Use a **heat-resistant surface** to avoid burn marks on your table.

The right amount of solder is crucial. Make sure the solder covers both the component and the PCB. The soldering iron must touch both the **component** and the **PCB** at the same time so that both are heated properly.

The following image shows typical solder joints: On the left, too little solder was used, while on the right, a ball forms – this happens when only the wire is heated. A perfect solder joint (middle) forms a smooth curve and the solder flows to the bottom of the PCB.



1.2 Resistors

Resistors are found in almost every circuit and are one of the three basic **passive components**, along with inductors and capacitors.

The job of a resistor is to **limit** the flow of current. The unit of resistance is the Ohm¹ (Ω). The relationship between current, voltage, and resistance is described by the well-known **Ohm's Law**:

$$U = R \cdot I$$
$$R = \frac{U}{I}$$

¹Named after the German physicist Georg Simon Ohm, born in 1789

The **resistance value** of a component can be read differently depending on its type: For SMD² components, the value is printed as a code, while for through-hole resistors, it's shown by color bands.

SMD resistors usually have a three-digit number like 471. The first two digits give the value, and the last digit shows the number of zeros. So, 471 means 47 followed by one zero, or 470Ω. A code like 683 means 68 followed by three zeros, or 68000Ω (68kΩ).

For through-hole resistors, the value is read using color bands. The following image shows how this works:

Farbe	1. Stelle	2. Stelle	(3. Stelle)	Multiplikator	Toleranz
Silber				x0,01	±10%
Gold				x0,1	±5%
Schwarz	0	0	0	x1	
Braun	1	1	1	x10	±1%
Rot	2	2	2	x100	±2%
Orange	3	3	3	x1.000	
Gelb	4	4	4	x10.000	
Grün	5	5	5	x100.000	±0,5%
Blau	6	6	6	x1.000.000	±0,25%
Violett	7	7	7	x10.000.000	±0,1%
Grau	8	8	8	x100.000.000	±0,05%
Weiß	9	9	9	x1.000.000.000	

4	7	x100	±5%	= 4k7Ω
6	8	0	x1	±0,1% = 680Ω

For SMD resistors, the resistance values can be identified based on their markings. The labeling works as follows: The first two digits indicate the value, and the third digit represents the number of zeros following the value. For example, a 220kΩ resistor is marked as "224," and a 470Ω resistor is labeled as "471".

Don't worry about damaging resistors when soldering—they are very heat-resistant and unlikely to get damaged. Also, it doesn't matter which way you solder them because resistors have no polarity. For the perfectionists: It looks neater if all the color bands face the same direction.

1.2.1 Exercises

A resistor can limit the flow of current, which is often useful when you don't want the current to go over a certain value.

The following two examples show how to calculate the resistance using Ohm's law. Ohm's law is:

$$U = R \cdot I \quad \text{or} \quad R = \frac{U}{I}$$

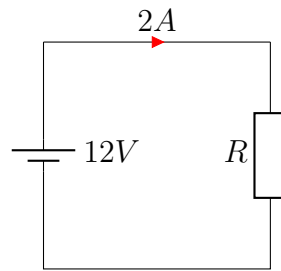
Example 1:

Let's say we have a circuit with a voltage of $U = 12V$ and a current of $I = 2A$. To calculate the resistance, we put the values into the formula:

$$R = \frac{U}{I} = \frac{12V}{2A} = 6\Omega$$

So, the resistance in this circuit is 6Ω.

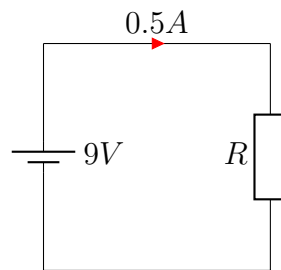
²Surface Mount Device

**Example 2:**

In another example, the voltage is $U = 9V$ and the current is $I = 0.5A$. The resistance is calculated as:

$$R = \frac{U}{I} = \frac{9V}{0.5A} = 18\Omega$$

So, the resistance in this circuit is 18Ω .



As you can see, it's easy to calculate the resistance when you know the voltage and current.

Calculation examples:

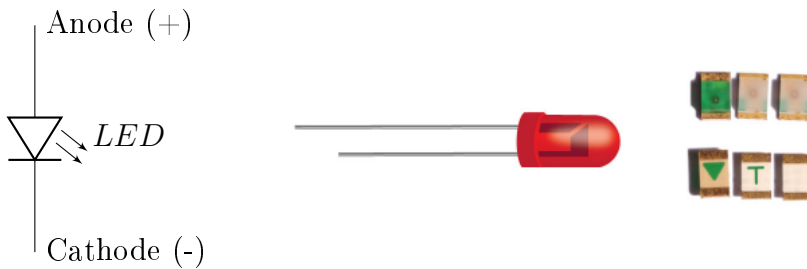
- A resistor with $R = 470\Omega$ is connected to a $5V$ power source. What is the maximum current? (Answer: $10.6mA$)
- A power supply of $24V$ should deliver a maximum current of $200mA$. What resistor do you need? (Answer: 120Ω)
- What is the voltage of a power supply if it provides a current of about $7mA$ through a 470Ω resistor? (Answer: $3.3V$)

1.3 Soldering LEDs

A light-emitting diode (LED) is an electronic component that emits **light** when current flows through it.

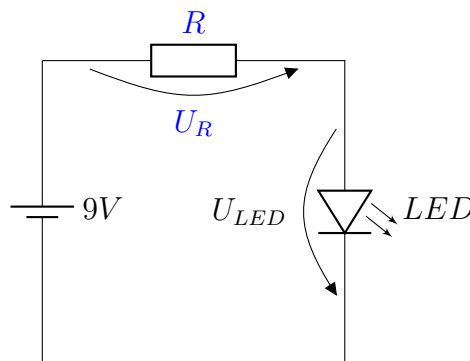
The correct **polarity** of the LED is crucial. If it is soldered onto the board the wrong way, it will not light up and might even be damaged.

The LED has two terminals: the **anode** (positive, +) and the **cathode** (negative, -). You can distinguish between them by the length of the legs. An easy way to remember is: "*Minus is always shorter*" – the shorter leg marks the cathode. For SMD LEDs, the cathode is often marked with a colored dot on the top or a diode symbol on the bottom.



An LED should never be operated without a **current-limiting resistor**. This resistor limits the maximum current flowing through the LED and protects it from overloading and potential damage. Always check the LED's datasheet to determine the correct values.

Example for calculating a current-limiting resistor:



Let's say we have an LED that lights up at a voltage of $U_{LED} = 2V$ and requires a maximum current of $I_{LED} = 20mA$. The supply voltage is $U_S = 9V$.

Since 2V are dropped across the LED, the remaining 7V are left for the resistor, as calculated by Kirchhoff's voltage law:

$$U_R = U_S - U_{LED} = 9V - 2V = 7V$$

The current flowing through the LED is the same as the current through the resistor. So, we can calculate the resistor's value using Ohm's law:

$$R = \frac{U_R}{I_{LED}} = \frac{7V}{0.02A} = 350\Omega$$

Thus, the required resistor is 350Ω . Since resistors come in standard values (called E-series), you often need to round up. The closest available value would be 360Ω .

You can also find a video on how to calculate the current-limiting resistor for LEDs on our YouTube channel:

<https://youtu.be/p0dt-6F-EZg>

1.3.1 Exercises

The answers are rounded to standard E-series resistor values!

- You have a low-current LED running at $I = 2mA$ and requiring a voltage of $2.5V$. You want to power it with $5V$. What current-limiting resistor do you need? (Answer: 1300Ω)
- A blue LED requires $4V$ and a current of $17mA$. What resistor should you use to power it with $12V$? (Answer: 470Ω)

2 LED alternating indicator SMD

2.1 Bill of Materials

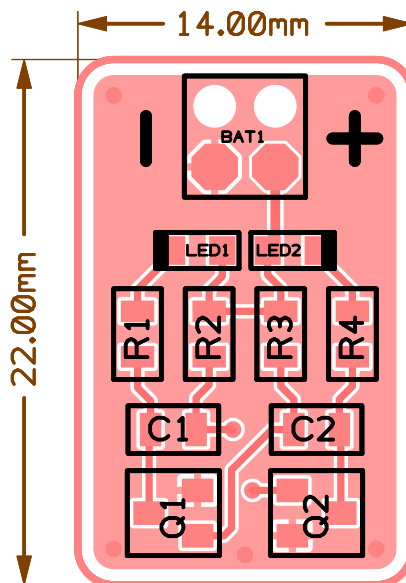
The following components are needed for assembly:

- PCB
- 2x 220k Resistor 0805 (R2 + R3)
- 2x 470R Resistor 0805 (R1 + R4)
- 2x NPN Transistor SOT23
- 2x Capacitor 0805
- 2x Red LED 0805
- 9V Battery Clip

The resistance values can be identified based on their markings. For SMD resistors, the labeling works as follows: the first two digits indicate the value, and the third digit represents the number of zeros following the value. For example, a $220k\Omega$ resistor is marked as "224," and a 470Ω resistor is labeled as "471."

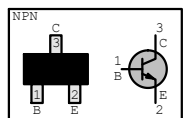
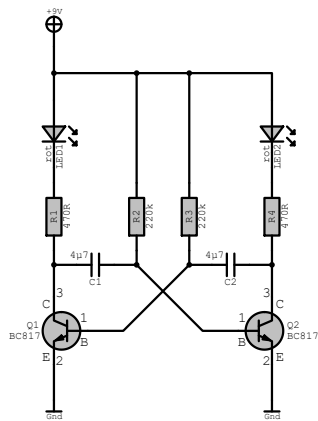
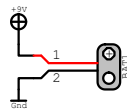
The critical components in this kit are the LEDs. Please pay close attention to their polarity before soldering them in. This is explained in the basic chapters.

All other components have no polarity and can be soldered in any direction. The orientation of the two transistors is self-explanatory.



2.2 Circuit Diagram

The circuit diagram for the kit is on the next page to ensure correct scaling.



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Gespeichert 10.09.2018, 15:23:01		Größe A4	Seite 1 von 1